AnimeGANv2 Video Inference Pipeline v4.1

This technical guide outlines the system architecture, component file inventory, environment requirements, and execution instructions for the optimized Linux-based AnimeGANv2 video transformation and caching pipeline (v4.1).

1. System Architecture & Requirements

- **OS Environment:** Linux (ext4 filesystem) with Conda environment manager.
- **Python Environment:** Python 3.10 inside Conda environment `animeenv`.
- **Core Dependencies:** PyTorch (CUDA acceleration enabled), OpenCV (`cv2`), FFmpeg 8.0+, NumPy, Argparse.
- **Model Weights:** Official generator architecture utilizing `face_paint_512_v2.pt`.

2. Complete File Inventory

File Path / Name	Description & Responsibility
`/home/admin1/aipipeline/batch_pipeline.sh`	Master orchestration script supporting dynamic `--threshold` parsing, video splitting, frame extraction, inference delegation, and final muxing.
`/home/admin1/aipipeline/inference_1080p.py`	Core inference engine with frame-diff caching logic, ETA tracking, and dynamic CLI argument parsing.
`/home/admin1/aipipeline/animegan2-pytorch/model.py`	Generator network architecture definition aligned with official weights.
`/home/admin1/aipipeline/animegan2-pytorch/weights/face_paint_512_v2.pt`	Pre-trained neural checkpoint weights.

3. Execution Instructions (v4.1)

To run the batch pipeline with custom frame-diff caching sensitivity (e.g., threshold set to `2.5`), execute the following command in your Linux terminal:

```
cd /home/admin1/aipipeline
./batch_pipeline.sh --threshold 2.5
```

To execute the inference script directly using the explicit Conda Python binary:

```
/home/admin1/miniconda/envs/animeenv/bin/python inference_1080p.py --threshold 2.5
```